

Document Map and Executive Summary

Media Reminder - Offline-First Adaptive Media Alarm

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2026-08-05

Contents

0.1 Document control

Field	Value
Document ID	MR-00
Version	1.0
Status	Approved baseline
Last updated	2026-08-05
Product owner	Mohamed Aslam Abdul
Package identifier	com.aslam.mediareminder
Purpose	Orient readers, declare the binding product baseline, and map every source-of-truth document.

Reading rule: This pack specifies a production-oriented Android application, not a promise that third-party devices will behave identically. Where Android or an OEM controls presentation, timing, sound, or permissions, the app provides transparent status and the strongest compliant fallback.

0.2 Document conventions

The words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are normative. A requirement ID is stable once published. Requirement IDs may be retired but **MUST NOT** be reassigned to a different meaning.

Term	Meaning
Reminder	A user-authored instruction that connects content, a schedule, and a presentation profile.
Occurrence	One calculated due instance of a reminder.
Alarm session	The bounded runtime state created when an occurrence is actively alerting.
Media asset	An app-owned local video, audio,

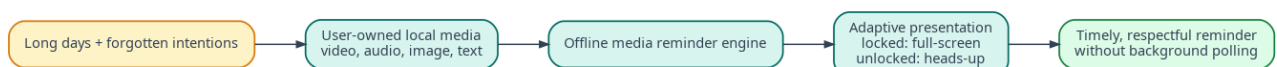
Term	Meaning
	image, or text item.
Profile	Reusable alert behavior such as Gentle, Standard, or Persistent.
Exact-alarm access	Android special app access that allows exact scheduling where the platform requires it.
Full-screen intent	Android notification mechanism for urgent, time-sensitive activity presentation. It is not a general overlay.
Heads-up notification	System-rendered high-priority notification shown temporarily over the current app when Android permits it.
Source of truth	The authoritative specification or local persisted record for a decision or state.

1 Executive summary

Media Reminder is an Android-first, offline-only application that lets a person import local media and schedule it as a reminder. The app adapts the interruption to device state:

- a locked or non-interactive device may wake into a full-screen alarm activity;
- an unlocked device uses Android's system heads-up notification and never forces a full-screen takeover;
- while Media Reminder itself is foregrounded, an in-app strip provides the least disruptive interaction.

The reminder does not autoplay its attached video merely because it is due. Alert sound and vibration may begin according to the selected profile, but the reminder media begins only after the user selects **Play**. The application is local-only: no account, server, advertising, telemetry, or Internet permission is required. It is scheduled by Android rather than by an always-running loop.



2 Binding baseline

Area	Approved baseline
Product name	Media Reminder
Initial platform	Android phones and foldables; tablets are supported with responsive layouts but are not the primary alarm surface
App framework	React Native 0.86.x, TypeScript, New Architecture
Reliability core	Kotlin native module, Android components, Room, DataStore,

Area	Approved baseline
	AlarmManager, Media3
SDK baseline	minSdk 26, compileSdk 37, targetSdk 36 for the first public release; retest and move target to 37 when required by distribution policy
Connectivity	No backend and no INTERNET permission in the production manifest
Persistence	App-owned files plus Room; system settings are discovered, not treated as app-owned data
Scheduler	One next due system alarm, recalculated after every relevant state change; no second-by-second polling and no resident idle service
Locked-device presentation	Full-screen intent/activity only when the device is locked or non-interactive and Android permits it
Unlocked-device presentation	High-importance heads-up notification; exact visual height is controlled by Android and the OEM
Backup	Versioned logical ZIP containing JSON records and media with SHA-256 checksums; never a raw database copy
Distribution	Signed APK and checksums through GitHub Releases; Play distribution remains optional
License	Apache License 2.0 for application source, excluding user-provided media and brand assets identified separately

3 Product boundaries

Media Reminder is suitable for personal duas and adhkar, study clips, exercise demonstrations, routines, language practice, and other rich-media reminders. It is **not** certified as a medical device, emergency warning system, safety-of-life alarm, or guaranteed exact-timing service. Android permissions, Doze policy, OEM modifications, device shutdown, force-stop, and user-disabled notification channels can prevent or alter an alert. The product **MUST** explain those states rather than claiming impossible certainty.

The application **MUST NOT**:

1. upload or analyze a person's media;
2. request overlay permission to draw over other apps;
3. ask for unrestricted battery optimization exemption as a default setup step;
4. keep a foreground service running while idle;

5. silently enable a reminder whose exact scheduling prerequisites are unavailable;
6. restore operating-system permissions, notification channel choices, or ringtone URIs from a backup as though they were portable app data.

4 Source-of-truth map

ID	Document	Authoritative for
MR-00	Document Map and Executive Summary	Scope map, precedence, fixed baseline
MR-01	Vision and Product Charter	Purpose, principles, goals, non-goals
MR-02	Product Requirements Document	Users, outcomes, scope, requirements, success criteria
MR-03	User Experience and Interaction Specification	Screen flows, adaptive alarm interaction, copy, states
MR-04	Visual Design System	Tokens, components, responsive behavior, motion
MR-05	Functional Feature Specification	Feature-level behavior and acceptance rules
MR-06	Android Alarm, Notification and Battery Specification	Native scheduling, alerts, permissions, lifecycle
MR-07	Technical Architecture	Components, dependency direction, deployment architecture
MR-08	Internal Module and Data Contracts	Typed boundaries, commands, events, error contracts
MR-09	Database and Local Storage Specification	Tables, relationships, migrations, files and transactions
MR-10	Backup, Export and Import Specification	Portable archive format and restore algorithm
MR-11	Edge Cases, Failure Recovery and Conflict Handling	Failure taxonomy, degradation and recovery rules
MR-12	Security, Privacy and Threat Model	Assets, trust boundaries, misuse controls, secure defaults
MR-13	Accessibility, Localization and Inclusive Design	WCAG-informed behavior, TalkBack,

ID	Document	Authoritative for
		scaling, RTL and copy
MR-14	Testing, Quality and Device Matrix	Test strategy, devices, release evidence and defect gates
MR-15	Performance, Battery and Operational Budgets	Quantitative budgets and measurement procedure
MR-16	Roadmap, Release and Success Metrics	Release increments, metrics and change control
MR-17	Architecture Decision Records	Binding architectural choices and their rationale
MR-18	Contribution Guide and Engineering Standards	Repository rules, code review and implementation quality
MR-19	AI Agent Guide and Master Loop Prompt	Safe AI-assisted development loop and stop conditions
MR-20	Release, Distribution, Portfolio and Maintenance Guide	Signing, APK publication, support and portfolio evidence
MR-21	Requirements Traceability and Acceptance Catalog	Requirement-to-test mapping and release acceptance
MR-22	Research Baseline and Official Sources	Current official platform facts and refresh policy

5 Requirement taxonomy

Requirement identifiers use a prefix and a sequence. The catalog in MR-21 is authoritative.

Prefix	Domain	Example
PRD	Product outcome or scope	PRD-012
UX	Interaction or presentation	UX-041
FUN	Feature behavior	FUN-103
AND	Android/native behavior	AND-057
DAT	Persistence and integrity	DAT-026
BKP	Backup/restore	BKP-044
SEC	Security/privacy	SEC-018
ACC	Accessibility/localization	ACC-021
NFR	Performance/battery/	NFR-034

Prefix	Domain	Example
	reliability	
REL	Release/distribution	REL-015

Every implemented requirement **MUST** have at least one verification method: automated test, instrumented test, manual test, static inspection, or measured evidence. A requirement is not complete because a screen exists; the mapped acceptance evidence must pass.

6 Product state model

The app separates four kinds of state:

1. **Intent state** - reminders, schedules, profiles, categories and user settings.
2. **Derived state** - next occurrence, notification content and calculated recurrence.
3. **Runtime state** - active alarm session, bounded ringing service, wake state and action idempotency.
4. **Platform state** - notification permission, exact-alarm access, channel importance, full-screen eligibility, battery policy and device lock state.

Only the first three are persisted as app state. Platform state is probed and displayed. This distinction prevents backup restore, migrations, and UI assumptions from overwriting choices owned by Android.

7 Release definition

A public v1 release is acceptable only when:

- all P0 requirements in MR-21 pass;
- there are no open critical or high-severity security, data-loss, alarm-action, or accessibility defects;
- locked, unlocked, Doze, reboot, timezone-change, force-stop explanation, denied-permission and backup recovery flows have been tested on the required device matrix;
- idle operation shows no resident foreground service and no periodic wake loop;
- import/export round trips preserve every supported logical record and media checksum;
- the APK is reproducibly built from a tagged commit, signed, checksum-published and accompanied by a privacy statement and known-limitations page;
- included screenshots and demo media are original, licensed, or intentionally synthetic.

8 Change control

A proposed change that affects alarm timing, permission scope, backup compatibility, native module contracts, file ownership, or public security posture **MUST** update the ADR log before implementation. Schema and backup changes **MUST** include migration and compatibility tests in the same pull request. Product copy changes that alter a promise about reliability **MUST** be reviewed against MR-06 and MR-11.

9 Implementation reading order

A developer or coding agent starts with MR-00, MR-17, MR-06, MR-07 and MR-09, then reads the relevant feature and UX documents. A designer starts with MR-01 through MR-05 and MR-13. A tester starts with MR-21, MR-14, MR-11 and MR-06. A release manager starts with MR-20, MR-14, MR-15 and MR-22.

9.1 Governance

This document is part of the **Media Reminder Source-of-Truth Pack v1.0**. In case of conflict, apply this precedence order: (1) explicit platform safety and permission rules, (2) Architecture Decision Records, (3) Android specification, (4) data and backup specifications, (5) PRD and feature specification, (6) UX and visual guidance, (7) implementation notes. Record any intentional deviation in the decision log before release.